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Primary Knee

Can Machine Learning Methods Produce Accurate and Easy-to-Use Preoperative Prediction Models of One-Year Improvements in Pain and Functioning After Knee Arthroplasty?



Alex H.S. Harris, PhD, MS ^{a, b, *}, Alfred C. Kuo, MD, PhD ^c, Thomas R. Bowe, PhD ^a,
Luisa Manfredi, JD, MPH ^a, Narlina F. Lalani, BA ^d, Nicholas J. Giori, MD, PhD ^{a, e}

^a Center for Innovation to Implementation, VA Palo Alto Healthcare System, Palo Alto, CA

^b Department of Surgery, Stanford–Surgical Policy Improvement Research and Education (S-SPIRE) Center, Stanford, CA

^c San Francisco Veterans Affairs Medical Center, University of California, San Francisco, CA

^d Center for Care Delivery and Outcomes Research, VA Minneapolis Healthcare System, Minneapolis, MN

^e Department of Orthopedic Surgery, Stanford University School of Medicine, Stanford, CA

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ABSTRACT

Background: Approximately 15%–20% of total knee arthroplasty (TKA) patients do not experience clinically meaningful improvements. We sought to compare the accuracy and parsimony of several machine learning strategies for developing predictive models of failing to experience minimal clinically important differences in patient-reported outcome measures (PROMs) 1 year after TKA.

Methods: Patients (N = 587) in 3 large Veteran Health Administration facilities completed PROMs before and 1 year after TKA (92% follow-up). Preoperative PROMs and electronic health record data were used to develop and validate models to predict failing to experience at least a minimal clinically important difference in Knee Injury and Osteoarthritis Outcome Score (KOOS) Total, KOOS JR, and KOOS subscales (Pain, Symptoms, Activities of Daily Living, Quality of Life, and recreation). Several machine learning strategies were used for model development. Ten-fold cross-validation and bootstrapping were used to produce measures of overall accuracy (C-statistic, Brier Score). The sensitivity and specificity of various predicted probability cut-points were examined.

Results: The most accurate models produced were for the Activities of Daily Living, Pain, Symptoms, and Quality of Life subscales of the KOOS (C-statistics 0.76, 0.72, 0.72, and 0.71, respectively). Strategies varied substantially in terms of the numbers of inputs required to achieve similar accuracy, with none being superior for all outcomes.

Conclusion: Models produced in this project provide estimates of patient-specific improvements in major outcomes 1 year after TKA. Integrating these models into clinical decision support, informed consent and shared decision making could improve patient selection, education, and satisfaction.

Level of Evidence: Level III, diagnostic study.

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For the great majority of end-stage knee osteoarthritis patients, elective total knee arthroplasty (TKA) is a life enhancing treatment. However, as many as 20% of TKA patients are dissatisfied with their long-term outcomes [1–3]. Although researchers have examined the associations between preoperative factors and improvements

in patient-reported outcome measures (PROMs) after TKA [4–10], with rare exceptions [11,12], these studies have not been designed to produce person-specific predictions of outcomes for new patients. The ability to preoperatively predict a patient's likelihood of experiencing clinically meaningful longer term improvements in PROMs after TKA—specifically identifying which patients are likely to be in the 20% who do not meaningfully benefit from the procedure—could substantially improve the quality of patient selection, personalized informed consent, shared decision making, and value-based reimbursement programs. However, developing models that are accurate and simple enough for routine clinical use is challenging.

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* Reprint requests: Alex H. S. Harris, PhD, MS, Center for Innovation to Implementation, VA Palo Alto Healthcare System (MC 151), 795 Willow Road, Menlo Park, CA 94025.

Fontana et al [12] recently compared 3 machine learning (ML) methods—least absolute shrinkage and selection operator (LASSO) regression, gradient boosting machines (GBM), and support vector machines—on a large single-institution joint replacement registry to predict which patients would fail to achieve a minimal clinically important difference (MCID) in 4 PROMs (36-Item Short Form Health Survey Mental [mental health component summary] and Physical Scales [physical health component summary], and Knee Injury and Osteoarthritis Outcome Score [KOOS], and KOOS Junior [KOOS JR]) 2 years after total knee replacement. They did not find that the ML methods differed from each other in terms of model accuracy. They were able to produce models restricted to the 3 preoperative PROMs as inputs (“Abbreviated 2”) that were reasonably accurate and well calibrated—C-statistics were 0.71, 0.88, and 0.71 for the physical health component summary, mental health component summary, and KOOS JR, respectively. However, these models require 46 items (3 baseline scales) as inputs which may impede adoption. Also, the JR version of KOOS may not be as clinically useful as the specific subscales (eg, Pain, Activities of Daily Living [ADL], Sports and Recreation) for the purposes of informed consent and shared decision making. A patient might be predicted to achieve an MCID in some domains and not others, granularity that is clouded by using the JR scale.

Therefore, the purpose of this study is to compare the accuracy and parsimony of several ML strategies for predicting whether TKA patients will achieve MCIDs in KOOS Total, KOOS JR, and KOOS subscales (Pain, Symptoms, ADL, Quality of Life [QoL], and Recreation) 1 year after TKA in Veteran Health Administration (VA) patients.

Materials and Methods

We conducted a prospective cohort study of patients undergoing elective primary TKA in 3 large, high-volume VA medical centers (Palo Alto, San Francisco, and Minneapolis). Revisions, bilateral TKA, and unicompartmental knee arthroplasties were not included. Patients were recruited by research staff during an orthopedic preoperative clinic visit within 30 days prior to their scheduled surgery. Patients were asked to complete preoperative questionnaire, to grant permission to link questionnaire data to their VA medical record data, and given permission to recontact them 1 year after surgery to complete a follow-up questionnaire. Patients who did not receive their originally scheduled TKA were removed from the cohort. Patients who underwent TKA were approached at their 1-year follow-up visit and/or by letter and phone to complete the follow-up questionnaire. Patients received \$25 for each completed questionnaire. This study was approved by the VA Central Institutional Review Board.

Questionnaires

The preoperative questionnaire included demographic information (gender, age, race/ethnicity, marital status, highest level of education completed, and employment status) [13] and the KOOS v1k1.01, allowing the calculation of the 10-item KOOS JR and the 5 KOOS subscales: Symptoms, Pain, ADL, Sports and Recreation Activities, and (knee-related) QoL [14,15]. Improvement was defined as the difference between the postoperative score and the preoperative score, with positive scores signifying improvement (−500 to 500 possible range for total KOOS Total score and −100 to 100 possible range for JR and subscale scores). Also included in the baseline questionnaire were the 19-item Hospital for Special Surgery: Knee Replacement Expectations Survey [16], the 4-item Alcohol Use Disorders Identification Test [17,18], the Patient Health Questionnaire-2, a common depression screening instrument [13], and the 5-item EuroQol 5-Dimension 5-Level Health Questionnaire, a measure of health-related QoL [19].

In addition to the KOOS, the 1-year follow-up questionnaire included the 4-item Self-Administered Patient Satisfaction Scale (SAPS) [20]. The SAPS contains 4 items: (1) satisfaction with results of surgery, (2) satisfaction with results of surgery for improving pain, (3) satisfaction with results of surgery for improving ability to do home or yard work, and (4) satisfaction with results of surgery for improving ability to do recreational activities. It is scored on a 4-point Likert scale with responses consisting of very satisfied (100 points), somewhat satisfied (75 points), somewhat dissatisfied (50 points), and very dissatisfied (25 points). The SAPS total score is the unweighted mean of the individual items, ranging from 25 to 100.

Medical Record Data

VA Corporate Data Warehouse was used to determine each patient's body mass index and if they had each comorbidity in the Elixhauser Comorbidity Index [21] recorded in their medical record in the 12 months prior to surgery.

Outcomes

We developed separate predictive models for the dichotomous outcomes of how likely a patient was to not experience an MCID on the KOOS Total score and each of the KOOS subscales. MCIDs were determined using Youden Index, an anchor-based method that identifies the magnitude of improvement that balances sensitivity and specificity in categorizing patients as being satisfied with their perceived improvement. As described elsewhere [22], we defined satisfaction as responding to all 4 SAPS items as at least somewhat satisfied.

Statistical Analysis

Our priorities for model development were accuracy, simplicity, and transparency to facilitate external validation and replication. Preoperative questionnaire and electronic health record data were used to develop and internally validate models to predict the odds that a patient would not experience at least an MCID in KOOS Total, KOOS JR, and KOOS subscales. Primary analyses for each PROM included the baseline value of the PROM and all other variables, not including the baseline values of the other PROMs. We did explore including all of the baseline PROMs, as well as the items rather than scale scores, but these approaches increased the number of required inputs and did not improve accuracy. Traditional and ML strategies were used for model development, including logistic regression, LASSO regression, GBM, and quadratic discriminant analysis (QDA).

These ML methods were chosen due to their unique strengths and weaknesses. Logistic regression is a traditional and transparent method, but requires manual variable selection and does not easily identify interactions and non-linearities. The main advantage of LASSO regression is automatic variable selection and more parsimonious models. However, like logistic regression, interactions and non-linearities need to be prespecified. GBM, an ensemble tree-based method, performs well in the presence of important interactions, and QDA performs well in the presence of nonlinear associations.

Ten-fold cross-validation and bootstrapping were used to produce 2 measures of overall accuracy: (1) Discrimination, quantified by the C-statistic, is the ability of a model to distinguish patients who experienced an outcome from those who do not [23]. In 10-fold cross-validation, 90% of the data is used to train the model and 10% is used for testing. This process is repeated until all 10 folds have been used for testing and the overall results are combined. In general, C-statistics can be interpreted as excellent (0.9–1), good

Table 1
Baseline Characteristics of Total Knee Replacement Cohort by Follow-Up Status.

Variable	Baseline Sample (N = 637)	Follow-Up (N = 587)	Lost to Follow-Up (N = 50)	P-Value ^a
Age (SD)	66.1 (8.3)	66.1 (8.2)	66.8 (9.1)	.67
BMI (SD)	33.3 (12.6)	33.4 (13.1)	32.6 (5.0)	.38
Female (%)	6.6	6.8	4.0	.44
Race/ethnicity (%)				.06
Caucasian	78.5	79.7	64.0	
African American	8.0	7.7	12.0	
Hispanic	9.6	9.2	14.0	
Asian	2.5	2.2	6.0	
Native American	3.6	3.9	0.0	
Marital status (%)				.12
Married	61.1	62.0	50.0	
Previously married	26.5	26.4	28.0	
Single	12.4	11.6	22.0	
Education (%)				.66
High school or less	27.1	26.8	30.0	
Some college	43.2	43.3	42.0	
College graduate	29.7	29.9	28.0	
Employment status (%)				.69
Full time	12.9	12.9	12.0	
Part time	7.1	7.3	4.0	
Unemployed/disabled	18.9	18.7	20.0	
Retired	61.1	61.1	64.0	
Congestive heart failure	4.2	4.4	2.0	.41
Valvular disease	4.1	4.4	2.0	.43
Peripheral vascular disease	5.2	5.1	6.0	.67
Hypertension	61.5	61.5	62.0	.94
Neurological disorders	4.4	4.4	4.0	.89
Chronic pulmonary disease	13.3	13.3	14.0	.88
Diabetes mellitus	26.1	25.9	28.0	.74
Hypothyroidism	6.9	7.3	2.0	.89
Renal failure	4.2	7.3	2.0	.17
Liver disease	4.9	5.1	2.0	.32
Solid tumor without metastasis	10.7	10.9	8.0	.52
Rheumatoid arthritis	3.8	3.7	4.0	.92
Weight loss	0.9	1.0	0.0	.47
Fluid and electrolyte disorders	4.6	4.4	6.0	.61
Deficiency anemia	7.5	7.3	10.0	.49
Alcohol use disorder	9.1	8.7	14.0	.21
Drug use disorder	5.4	5.6	4.0	.62
Depression	22.8	23.5	14.0	.12
AUDIT-C Total (SD)	2.1 (2.2)	2.1 (2.3)	1.7 (2.1)	.22
PHQ Total (SD)	1.8 (1.9)	1.8 (1.9)	2.1 (2.2)	.27

SD, standard deviation; BMI, body mass index; AUDIT-C, Alcohol Use Disorders Identification Test; PHQ, Patient Health Questionnaire.

^a P-values from tests (t-tests, chi-squared, tests of proportions) comparing completers from those who did not provide follow-up data.

(0.8–0.89), fair (0.7–0.79), poor (0.6–0.69), or fail/no discriminatory capacity (0.5–0.59) [24–26]. (2) Calibration refers to the association between predicted and observed outcomes across the entire range of the data. Calibration is considered good if prevalence of outcomes tracks monotonically with the predicted probabilities from the model. Calibration was primarily assessed by visually inspecting plots of observed vs predicted outcomes. To provide a quantitative assessment of calibration, we used the Brier score which is analogous to the sum of squared error in regression analysis in that smaller values signify better calibration [27]. The Brier score ranges from 0, signifying perfect calibration, to 1, signifying the worst possible calibration. Brier score can be used to compare the quality

of calibration between models but standards have not been proposed for values that signify good or poor calibration. Finally, because end users of these models may want to select a threshold of predicted probability for not achieving an MCID that will trigger a shared decision making or other preoperative decision process, the sensitivity and specificity of various predicted probability cut-points were examined.

Results

A total of 637 patients met eligibility criteria and provided baseline data, of whom 587 (92.2%) provided follow-up data 11–15

Table 2
Baseline PROs of Total Knee Replacement Cohort by Follow-Up Status.

Variable	Baseline (N = 637)	Follow-Up (N = 587)	No Follow-Up (N = 50)	P-Value ^a
KOOS Total	180.1 (71.9)	181.7 (70.7)	161.0 (83.4)	.09
KOOS Pain	43.9 (16.4)	44.1 (16.1)	41.2 (19.6)	.31
KOOS Symptoms	49.9 (19.1)	50.3 (18.9)	45.1 (20.1)	.08
KOOS ADL	47.5 (17.7)	47.7 (17.4)	44.4 (20.6)	.27
KOOS Recreation	17.6 (19.6)	18.1 (19.7)	11.9 (17.0)	.02
KOOS Quality of Life	21.4 (16.0)	21.6 (12.7)	18.4 (19.6)	.25
KOOS JR	45.8 (13.0)	46.1 (12.7)	43.3 (16.2)	.26

KOOS, Knee Injury and Osteoarthritis Outcome Score; ADL, ADL, activities of daily living; PROs, patient-reported outcomes.

^a P-value pertains to t-test comparing the follow-up to no follow-up samples.

Table 3
Improvements in Patient-Reported Outcomes of TKA Cohort (N = 587).

Variable	Follow-Up, Mean (SD)	Improvement, Mean (SD)	MCID ^a
KOOS Total	344.3 (95.0)	163.3 (96.9)	91.8
KOOS Pain	82.1 (18.2)	37.9 (20.4)	25.0
KOOS Symptoms	79.5 (17.2)	29.2 (22.1)	14.3
KOOS ADL	82.4 (18.2)	34.6 (20.5)	24.6
KOOS Recreation	51.8 (30.9)	34.1 (31.1)	17.5
KOOS Quality of Life	56.9 (26.3)	35.2 (26.9)	12.5
KOOS JR	76.8 (16.6)	30.7 (16.3)	20.8

SD, standard deviation; MCID, minimal clinically important difference; KOOS, Knee Injury and Osteoarthritis Outcome Score; ADL, ADL, activities of daily living.

^a Details of MCID calculation presented in Kuo et al (2020) [22].

months after their operation. Patients averaged 66.1 (8.3) years of age and were predominantly male (93.4%) and Caucasian (78.5%; Table 1). The sample of patients who provided follow-up data was similar to those who were lost to follow-up (N = 50), in terms of demographics and comorbidities (Table 1). The baseline values of the PROMs of the sample who provided follow-up data did not statistically differ from those who were lost to follow-up, except for the KOOS Recreation subscale (Table 2).

For the 587 patients who provided follow-up PROM data, improvements on the KOOS Total, KOOS subscales, and KOOS JR varied widely within each scale (Table 3, Fig. 1, and Supplemental File 1). The models with the best discrimination were for the KOOS ADL subscale (logistic; C-statistic 0.76), Pain subscale (LASSO, GBM, QDA; C-statistic 0.72), Symptoms subscale (LASSO; C-statistic 0.72), QoL subscale (GBM; C-statistic 0.71), and KOOS Total (QDA; C-statistic 0.66; Table 4). Discrimination for the KOOS JR and Recreation subscales were very poor (≤ 0.60) for all models. Calibration, as quantified by the Brier score, was very similar for all outcomes (0.14–0.19), except the Recreation subscale which had poor calibration.

Calibration plots show general good correspondence between predicted risk and prevalence of outcomes, especially for the models with better discrimination (Fig. 2 and Supplemental File 2).

The modeling strategies differed in terms of the numbers of inputs required to achieve similar accuracy, with none being superior for all outcomes. We attempted to identify models with the best balance of discrimination, calibration, and number of inputs (italicized in Table 4). For the KOOS Total, Pain, and Symptom subscales, QDA produced models that were superior or equal to other approaches on all 3 criteria. These models included only the baseline value of the PROM (varying in the number of items needed to calculate), plus indicators for 2 other variables: (1) whether the patient had a drug use disorder diagnosis in the medical record and (2) whether the patient indicated they worked full time. For the KOOS ADL subscale, the LASSO regression was slightly less accurate than logistic regression (C-statistic 0.72 vs 0.76) but had the same calibration (Brier = 0.17) and required far fewer inputs (27 vs 80). For the KOOS QoL subscale, the GBM regression was the most accurate and calibrated, but requires 53 inputs. We are developing on-line calculators to allow clinicians and patients to use these models to estimate the probability of experiencing MCIDs from elective TKA for all of the outcomes.

The sensitivity and specificity of various predicted probability cut-points for the most accurate models are presented in Supplemental File 3. Lower thresholds of predicted probability of not exceeding the MCID (eg, ≥ 0.10) result in more people being predicted to not achieve the MCID and higher sensitivity (ie, a greater percent of people who do not achieve the MCID will be correctly predicted) but lower specificity. Conversely, higher thresholds of predicted probability for exceeding the MCID (eg, ≥ 0.50) result in fewer people being predicted to not achieve the MCID (lower sensitivity) with higher specificity. The table presented in Supplemental File 3 allows selection of a threshold that fits a given context and set of priorities. For example, using a predicted

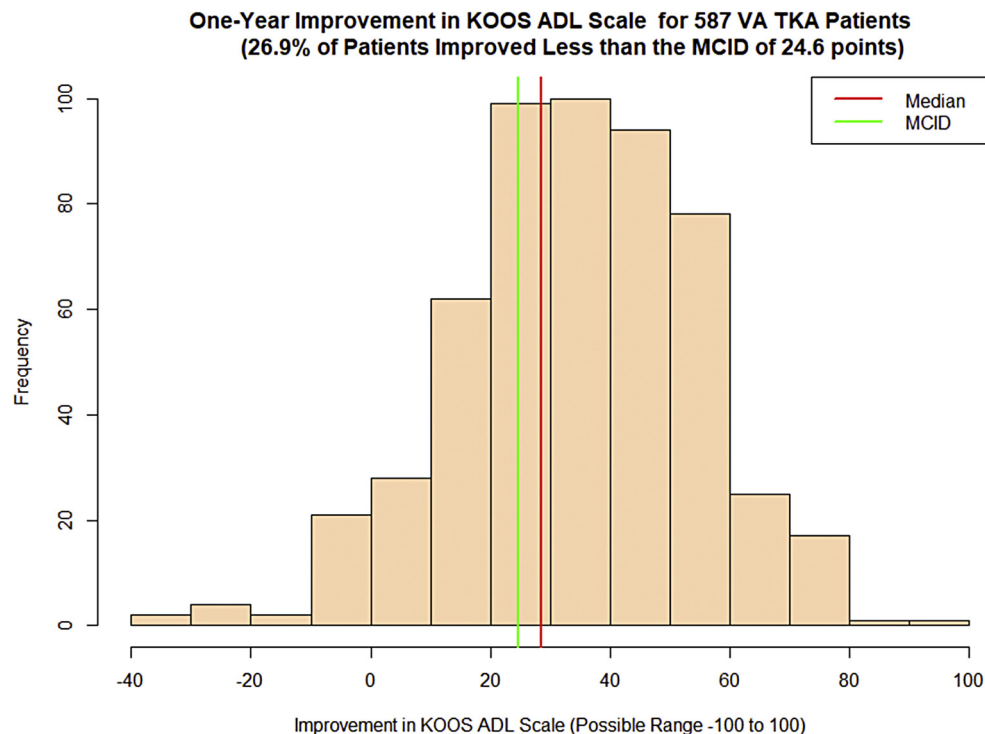


Fig. 1. One year improvement in KOOS ADL scale for 587 TKA patients. KOOS, Knee Injury and Osteoarthritis Outcome Score; ADL, activities of daily living; TKA, total knee arthroplasty; MCID, minimal clinically important difference.

Table 4
Discrimination, Calibration, and Number of Variables Selected for Different Modeling Strategies.

Domain	Logistic			LASSO			GBM			QDA		
	C-Stat	Brier	No. of Items	C-Stat	Brier	No. of Items	C-Stat	Brier	No. of Items	C-Stat	Brier	No. of Items
Total	0.64	0.17	106	0.62	0.17	98	0.63	0.16	87	0.66	0.16	43
Pain	0.71	0.17	72	0.72	0.16	59	0.72	0.16	64	0.72	0.15	9
Symptoms	0.68	0.19	68	0.72	0.17	22	0.71	0.17	41	0.71	0.17	7
ADL	0.76	0.17	80	0.72	0.17	27	0.71	0.17	34	0.63	0.20	19
Recreation	0.45	0.28	68	0.46	0.27	65	0.52	0.24	52	0.45	0.31	7
QoL	0.67	0.15	67	0.66	0.15	61	0.71	0.14	53	0.62	0.16	6
Junior	0.56	0.22	73	0.56	0.22	52	0.47	0.24	66	0.60	0.20	12

LASSO, least absolute shrinkage and selection operator; GBM, gradient boosting machines; QDA, quadratic discriminant analysis; ADL, activities of daily living; QoL, quality of life. Models with the best balance of accuracy and simplicity of inputs are italicized.

probability threshold of 0.4 for the KOOS Symptom Scale, that is, assuming that patients above that threshold will not achieve improvement by the MCID, will yield a sensitivity of 0.52 and specificity of 0.80.

Discussion

The ability to accurately predict which patients are unlikely to receive clinically important benefits from elective TKA could substantially enhance the quality of personalized informed consent, shared decision making, and value-based reimbursement programs. In this project, we compared modeling strategies with the goal of producing predictive models that are both accurate and simple enough for widespread use. We were able to produce models with C-statistics greater than 0.70 and good calibration for KOOS Pain, Symptoms, ADL, and QoL subscales, but not for the KOOS Total score, JR, or Recreation subscales. These models are being incorporated into an online tool that can be accessed by anyone. We caution potential users, however, that the models have been developed on veteran patients cared for in the VA healthcare system. External validation would be required to determine whether these models are applicable to the general arthroplasty population.

In terms of balancing accuracy, number of inputs, and transparency, we found that no modeling approach was superior for all outcomes. QDA was able to produce models with better or similar

accuracy with far fewer inputs. If accuracy and simplicity are the most important design criteria for the predictive model, then the QDA models are excellent for several outcomes. One needs to understand, however, that these are prediction models, and not explanatory models. The coefficients cannot be causally interpreted. Although the presence of a drug use disorder diagnosis and full-time work may have a causal relationship with outcomes, it is far more likely that these variables act as proxies for a large number of causally related variables that are associated with drug use and employment. These causally related variables may or may not have been included as candidate predictors in our analysis. Thus, if a patient who is unemployed when presenting for surgery suddenly gets a job right before surgery, the actual improvement in outcomes following surgery is unlikely to be boosted much as the actual causal underlying medical and social conditions that associate with unemployment may not have changed substantially for that particular patient.

Unlike Fontana et al [12], we were unable to produce an accurate prediction model for the KOOS JR even by using a more extensive group of inputs. We were able to produce accurate models for most of the KOOS subscales, in some cases with as few as 7 inputs. However, to get predictions for all of the KOOS total score and subscales, users will have to enter all 40 items of the KOOS plus at least 2 other items. This is the same input burden as the Fontana et al models, except for a more diverse set of outcomes.

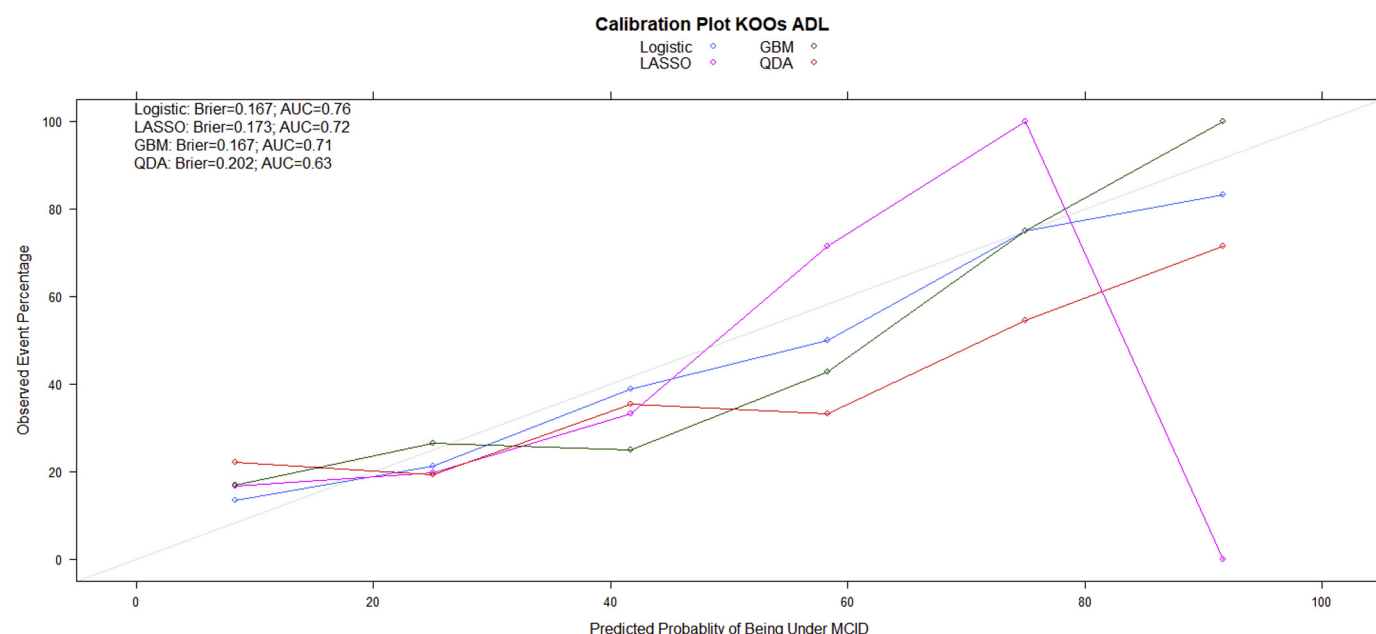


Fig. 2. Calibration plot KOOS ADL. KOOS, Knee Injury and Osteoarthritis Outcome Score; ADL, activities of daily living; MCID, minimal clinically important difference; QDA, quadratic discriminant analysis; LASSO, least absolute shrinkage and selection operator; GBM, gradient boosting machines.

The development and internal validation of prediction models is just the beginning. As we have described elsewhere, the path from accurate predictions to real-world benefits for patients and healthcare systems is long and demanding [28]. First, although these models may be accurate for use in the largest integrated healthcare system in the United States, efforts should still be made to improve their accuracy. They should also be externally validated before use in other settings. Second, the decision to proceed with elective TKA should be informed by balancing the risks against the benefits. Although prediction models of short-term mortality and complications of TKA have been developed by our team and others [26,29–31], developing calculators and clinical decision support systems that facilitate the surgical decision making process of balancing predicted risks against predicted benefits in a useable and interpretable interface has not yet been achieved. Third, the value of packaging the information produced by these models into preoperative processes, such as personalized informed consent or shared decision making, needs to be rigorously tested.

The outputs of these models are predicted probabilities, ranging from 0 to 1, of failing to achieve an MCID. Because it could be useful to establish probability thresholds that may trigger a decision to deny surgery or enhance the shared decision-making process, we have provided the sensitivity and specificity of categorizing a patient as not likely to achieve an improvement greater than the MCID throughout the range of possible cut-points. The optimal balance of sensitivity and specificity depends on one's priorities and preferences, which may differ between and among surgeons and patients. We cannot say which threshold is optimal for a particular decision. We have presented the data so those designing clinical processes and decisions based on thresholds can understand and account for these tradeoffs.

Conclusions

In summary, we have developed predictive models for failing to experience meaning improvements in patient-reported outcomes using various ML techniques for veteran patients cared for at VA healthcare facilities. These models have been internally validated and achieve fair metrics of discrimination for some, but not all, of the patient-reported outcomes collected. Continuing to improve predictive accuracy of these models, perhaps with the inclusion of novel biometric or patient-generated inputs, remains an important goal. The models reported here are freely available from the authors for testing in other healthcare systems and for overall model improvement, as well as to integrate into care processes to improve patient education, selection, and satisfaction.

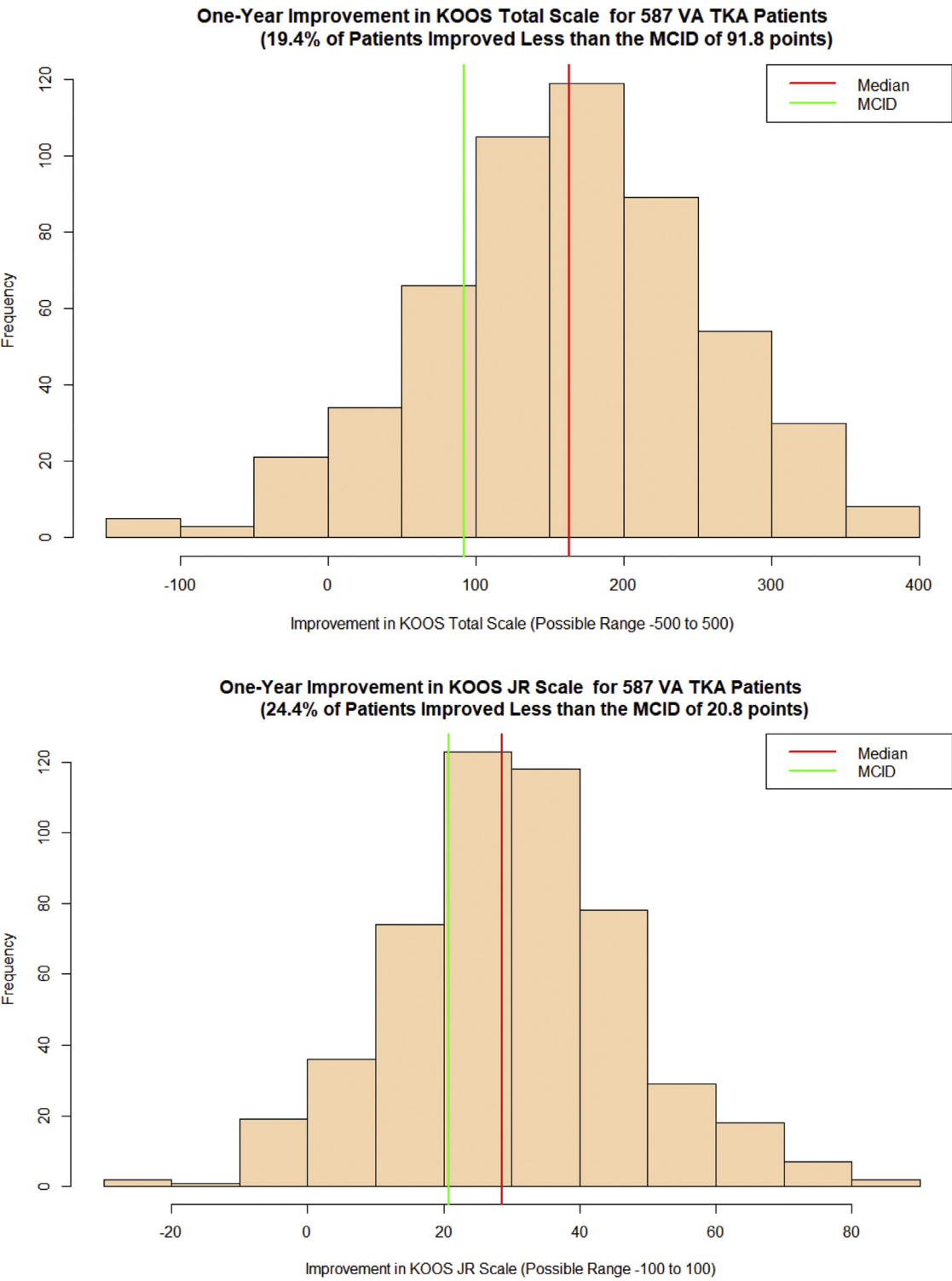
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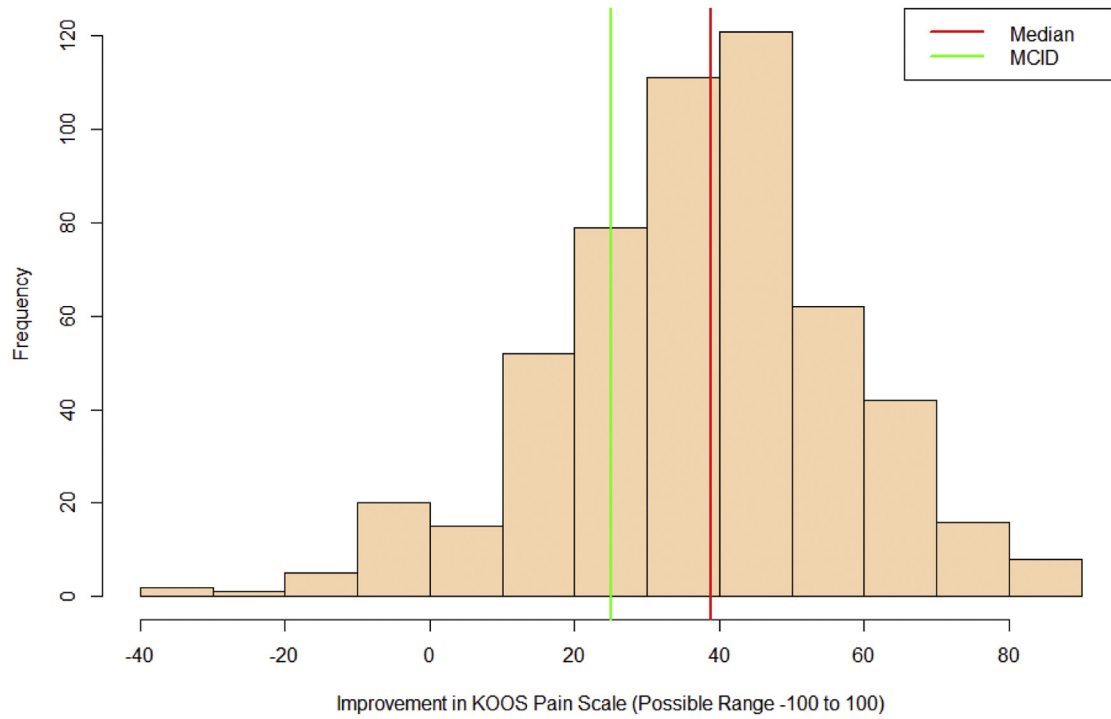
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Appendix

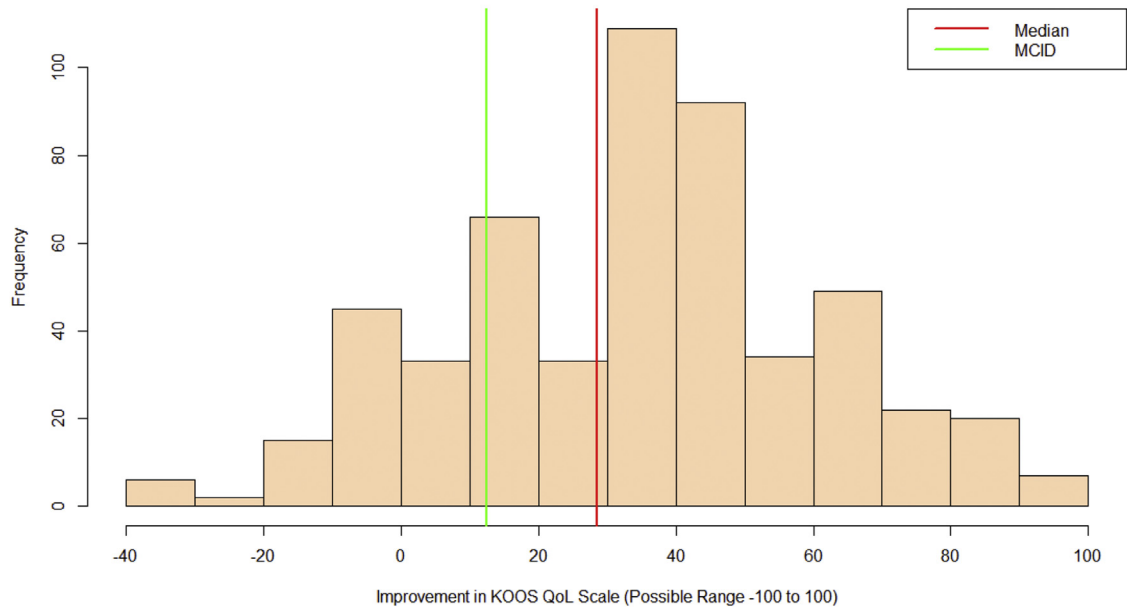


Supplemental File 1. Distribution figures for improvements in KOOS Total and Subscales. KOOS, Knee Injury and Osteoarthritis Outcome Score; VA, Veteran Health Administration; TKA, total knee arthroplasty; MCID, minimum clinically important difference.

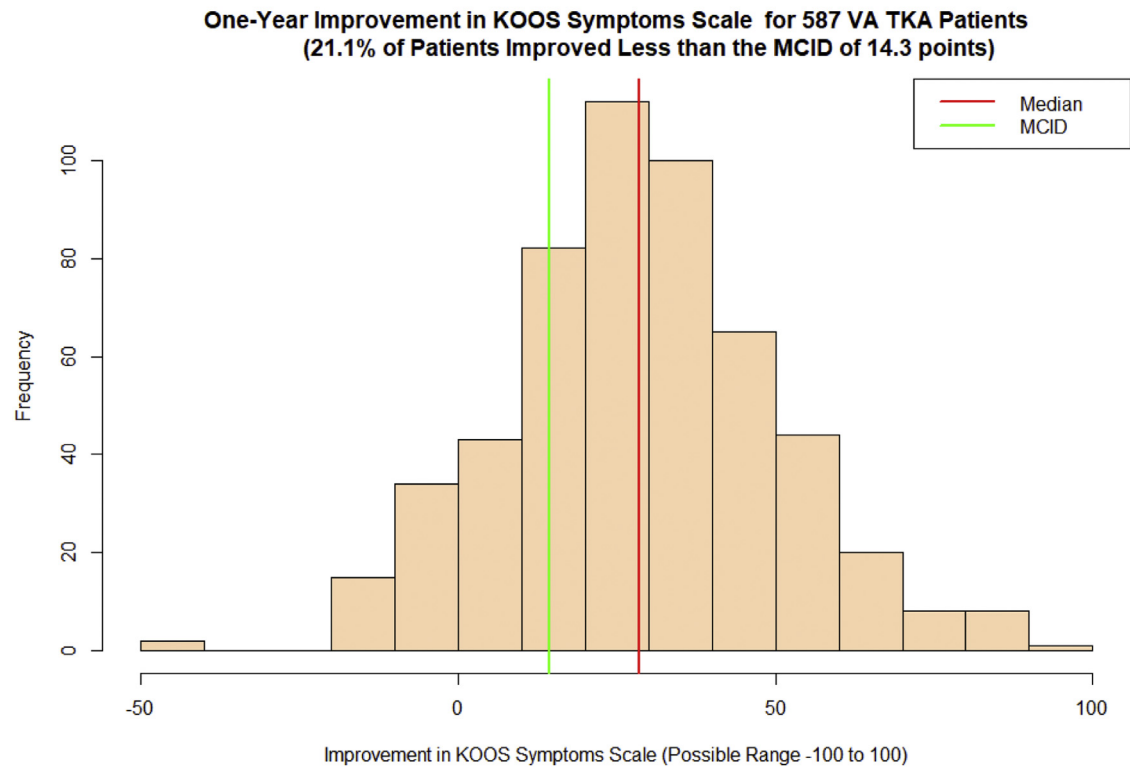
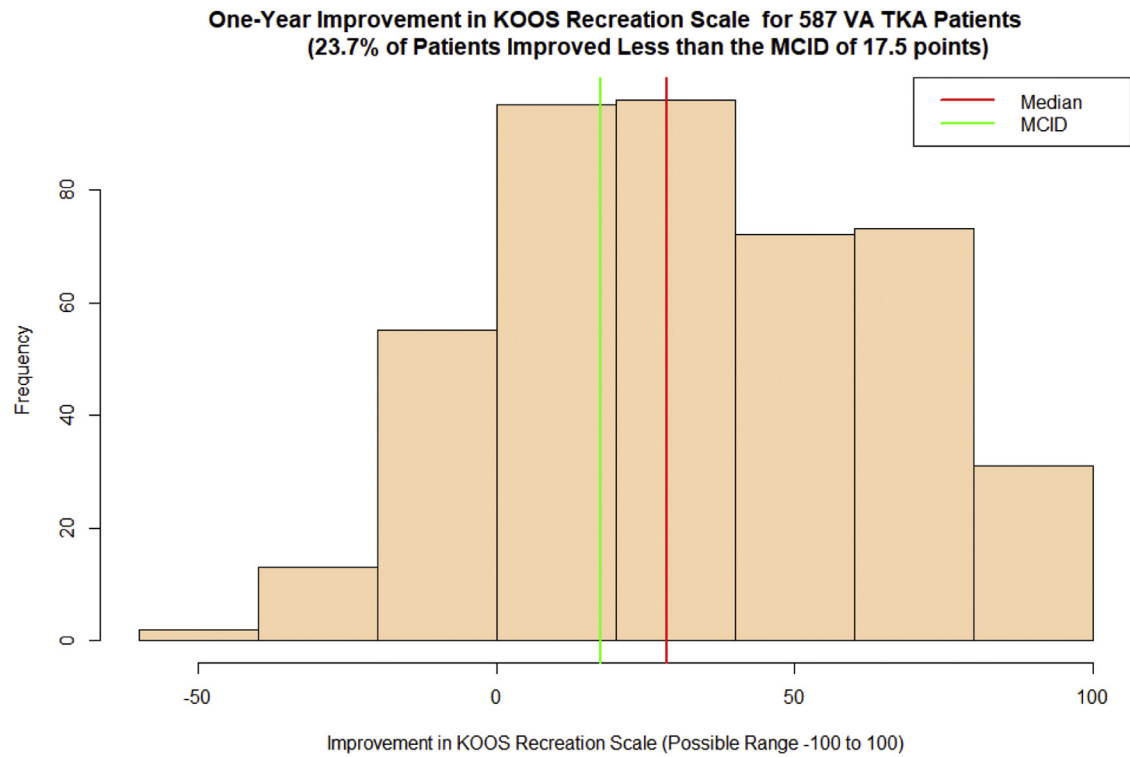
**One-Year Improvement in KOOS Pain Scale for 587 VA TKA Patients
(20.3% of Patients Improved Less than the MCID of 25.0 points)**



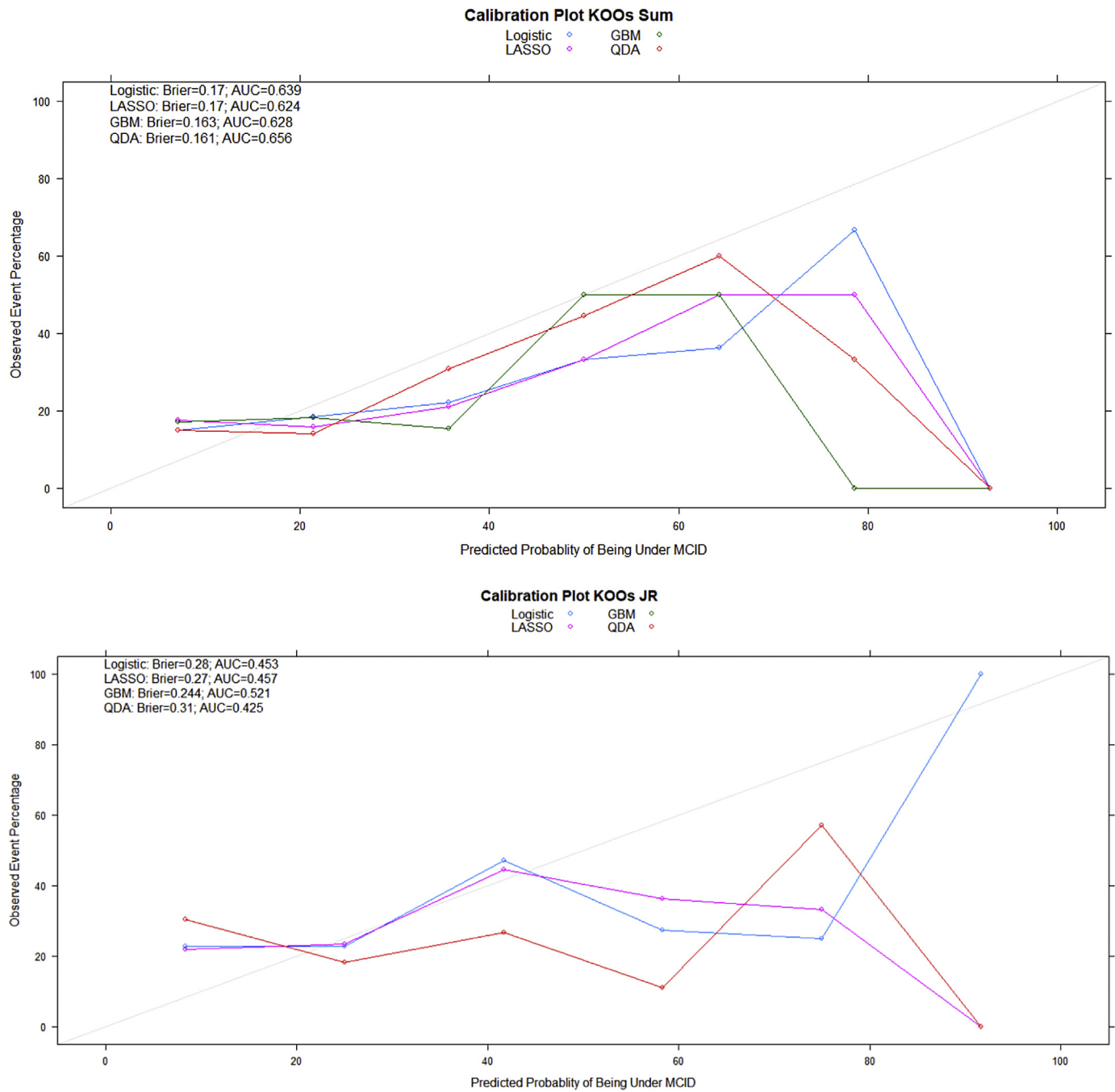
**One-Year Improvement in KOOS QoL Scale for 587 VA TKA Patients
(17.2% of Patients Improved Less than the MCID of 12.5 points)**



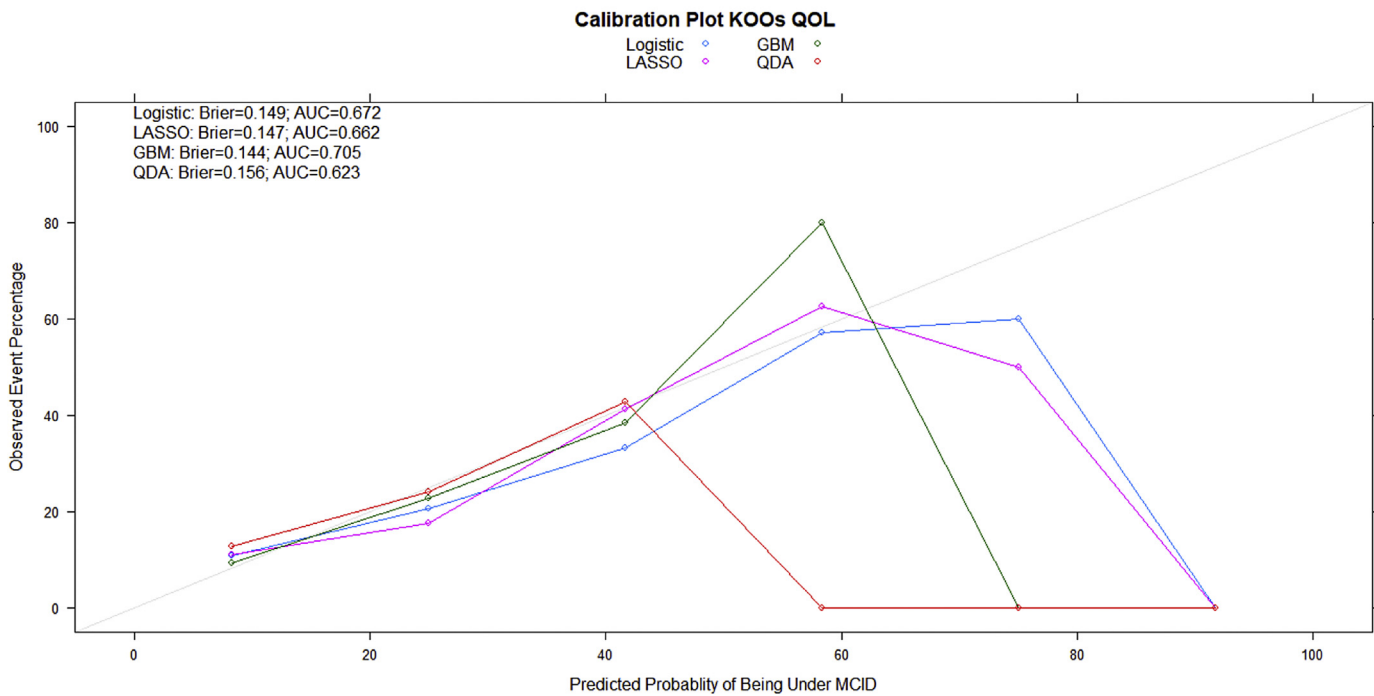
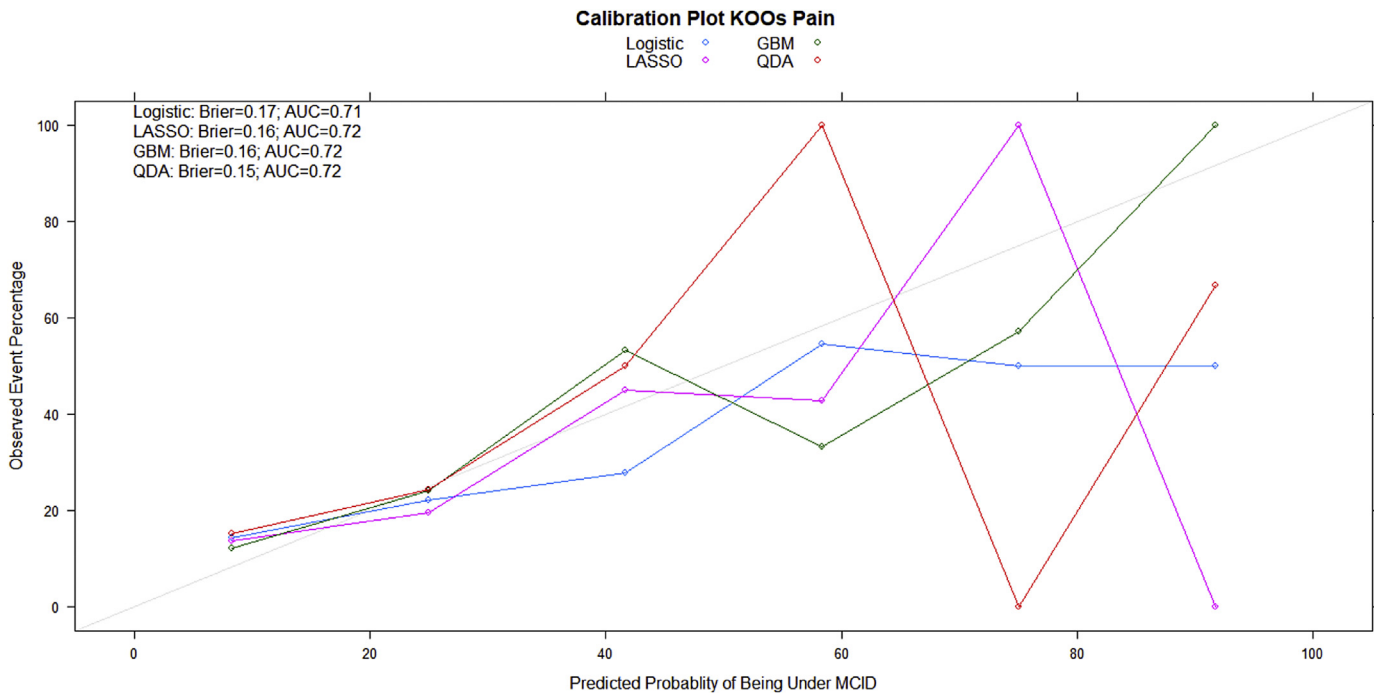
Supplemental File 1. (continued).



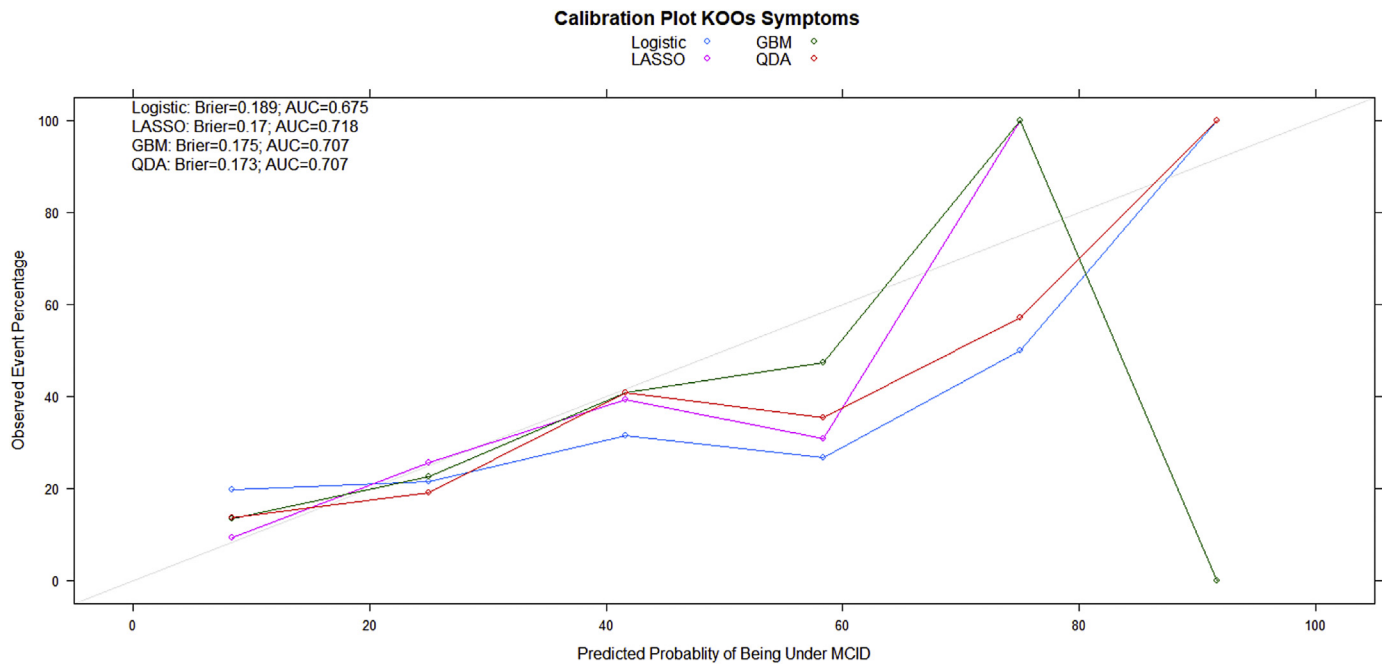
Supplemental File 1. (continued).



Supplemental File 2. Calibration plots for KOOS Total and Subscales. KOOS, Knee Injury and Osteoarthritis Outcome Score; LASSO, least absolute shrinkage and selection operator; GBM, gradient boosting machines; QDA, quadratic discriminant analysis; AUC, area under the curve; QoL, quality of life.



Supplemental File 2. (continued).



Supplemental File 2. (continued).

Supplemental File 3

Sensitivity and Specificity of Predicted Probability Cut-Points for Not Achieving the MCIDs.

Predicted Probability Threshold (Value Above Which People Are Expected to NOT Meet the MCID)	KOOS Total Sensitivity	KOOS Total Specificity	KOOS Pain Sensitivity	KOOS Pain Specificity	KOOS Symptoms Sensitivity	KOOS Symptoms Specificity	KOOS ADL Sensitivity	KOOS ADL Specificity	KOOS QoL Sensitivity	KOOS QoL Specificity
0.1	1	0	0.81	0.37	0.92	0.27	1	0.06	0.92	0.17
0.2	0.67	0.61	0.59	0.81	0.79	0.5	0.8	0.33	0.68	0.61
0.3	0.5	0.77	0.28	0.95	0.65	0.66	0.66	0.67	0.39	0.87
0.4	0.27	0.91	0.15	0.99	0.52	0.8	0.51	0.82	0.25	0.94
0.5	0.23	0.95	0.09	0.99	0.36	0.86	0.32	0.96	0.14	0.98
0.6	0.06	0.93	0.09	0.99	0.23	0.92	0.14	1	0.03	0.99

KOOS, Knee Injury and Osteoarthritis Outcome Score; MCID, minimum clinically important difference; ADL, activities of daily living; QoL, quality of life.